Phase 6: KiCad Project Structure

Marklin WiFi AC Train Controller

2026-03-05

Test Project: KiCad Project Structure

Project Details

Field	Value
Project name	marklin-wifi-ctrl
Author	PA Nilsson
Company	ZID AB
Revision	0.1
KiCad version	9.0.7
File format version	20250114

Component Import

Three components were imported via easyeda2kicad --full. Each import downloads a symbol, footprint, and 3D model:

Component	LCSC	Symbol	Footprint	3D Model
ESP32-C3-MINI-1-N4	C2838502	ESP32-C3-MINI-1-N4	WIFIM-SMD_ESP32-C3-MINI-1	WIFIM-SMD_ESP32-C3-MINI-1 (.wrl/.step)
EL357N(C)(TA)-G	C29981	EL357N(C)(TA)-G	OPTO-SMD-4_L4.4-W4.1-P2.54-LS7.0-TL	OPTO-SMD-4P_L4.1-W4.4-H2.0-LS7.0-P2.54 (.wrl/.step)
MOC3021S-TA1	C115465	MOC3021S-TA1	SMD-6_L7.3-W6.5-P2.54-LS10.2-BL	SMD-6_L7.3-W6.5-H3.6-LS10.16-P2.54 (.wrl/.step)

After import, the libraries were cleaned to contain only these 3 components (the shared easyeda2kicad output directory contained 27 symbols and 11 footprints from prior imports).

Project-specific libraries named after the project:

- marklin-wifi-ctrl.kicad_sym - 3 symbols
- marklin-wifi-ctrl.pretty/ - 3 footprints
- marklin-wifi-ctrl.3dshapes/ - 3 models (6 files: .wrl + .step each)

All internal references updated: symbols reference marklin-wifi-ctrl:<footprint>, footprints reference ${KIPRJMOD}/marklin-wifi-ctrl.3dshapes/<model>.wrl.

Project File Structure

```
marklin-wifi-ctrl/
  marklin-wifi-ctrl.kicad_pro     # Project file
  marklin-wifi-ctrl.kicad_sch     # Root schematic (hierarchy overview)
  connectors.kicad_sch            # Block 9: Track input + motor output
  power_supply.kicad_sch          # Block 1: 24VAC to 3.3VDC
  esp32_c3.kicad_sch              # Block 2: ESP32-C3-MINI-1 module
  zero_crossing.kicad_sch         # Block 3: Zero-cross detector
  voltage_sense.kicad_sch         # Block 6: AC voltage measurement
  hw_interlock.kicad_sch          # Block 4: NAND interlock
  triac_drive.kicad_sch           # Block 5: MOC3021S + BT136 (x2)
  status_led.kicad_sch            # Block 7: Status LED
  usb_prog.kicad_sch              # Block 8: USB-C programming
  sym-lib-table                   # Symbol library table (registers project libs)
  fp-lib-table                    # Footprint library table (registers project libs)
  marklin-wifi-ctrl.kicad_sym     # Imported symbol library (3 symbols)
  marklin-wifi-ctrl.pretty/       # Imported footprint library (3 footprints)
  marklin-wifi-ctrl.3dshapes/     # Imported 3D models (3 models)
```

Root Schematic Layout

The root schematic (A3 paper) arranges the 9 hierarchical sheet symbols in a 3x3 grid:

Row	Left	Center	Right
Row 1 (top)	Connectors	Power Supply	ESP32-C3 MCU
Row 2 (mid)	Zero-Crossing	Voltage Sensing	Hardware Interlock
Row 3 (bot)	TRIAC Drive	Status LED	USB Programming

Signal flow is left-to-right (AC domain to DC domain) and top-to-bottom (power to output).

Verification

- Project loads successfully in KiCad 9.0.7
- All 10 sheets (1 root + 9 children) export to SVG without errors
- Title blocks show correct project name, author, company, and revision